

Towards a MATSim Scenario for a Data-Scarce Megacity Dhaka, Bangladesh

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MOTIVATION

Agent-based models need data that Dhaka does not have in usable form

Census locked in a PDF

Ward-level data exists only as a 1,668-page report with no machine-readable file.

Survey without Origin Destinations

The household travel survey records no trip destination coordinates, and no O-D matrix exists.

No transit feed

No official GTFS exists for Dhaka or Bangladesh, and much of the bus network is informal.

Aim: build a reliable agent-based model anyway, and share a workflow other low- and middle-income cities can reuse.

Related Works

Data-rich Settings

- Eqasim for Île-de-France (Hörl & Balać 2021)
- MATSim-NYC (Chow et al. 2020)
- Washington, D.C. mobility twin (Ruan et al. 2025)
- Ústí nad Labem (Pereira et al. 2022)
All rely on GTFS feeds, building registries and fine census tracts.

Data-scarce Settings

- Greater Jakarta: 30 million people synthesized with IPU (Kagho et al. 2020)
- Dhaka BRT model in MATSim (Zannat et al. 2024a)
The Dhaka model teleported public and non-motorized transport, as no transit route data existed.

This study: route public transport on the real network.

STUDY AREA

A megacity that moves on foot and by rickshaw

10,295,786

residents in Dhaka North and South city corporations (BBS 2022)

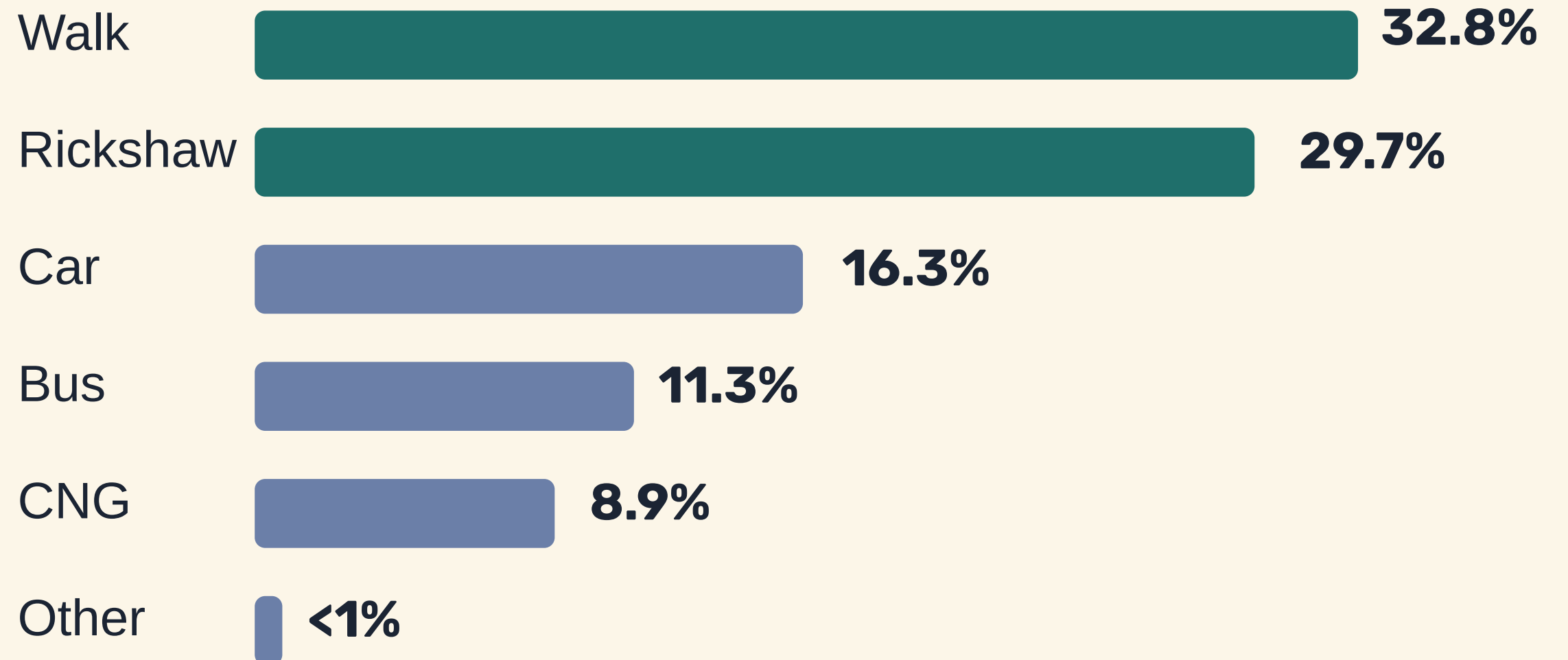
2.5 million

more in the adjoining Savar and Keraniganj upazilas

130 + 2

wards (codes such as S07, N54) plus two suburban zones

Modal split, weighted household travel survey



Five data sources, four stages, one MATSim scenario

Input Preparation

Clean the survey, parse census tables, unify ward IDs, classify buildings

Activity locations

Place home, work, education and secondary activities; build daily plans

Population Synthesis

Reweight survey households with IPU, integerize with TRS, assign wards

Transit supply

Build GTFS from route documents; map with pt2matsim

Filling each data gap

Census microsample	Large weighted household sample	IPU raking of the travel survey sample
Fine-grained census	Structured microdata or tables	1,668-page PDF, parsed and cross-validated
Household travel survey	One respondent per household	Full roster of 52,672 households
Commuting O-D matrix	Municipality-to-municipality flows	Distance-distribution location matching
Facility registry	Address and enterprise censuses	<1% of OSM buildings typed; add OSM POIs
Transit schedule	Official GTFS feed	Rebuilt from DTCA route documents

Building the population

Targets

Ward households, sex and
age from the BBS 2022
Community Report

IPU fitting

Reweights survey
households to match
household and person
totals together

TRS

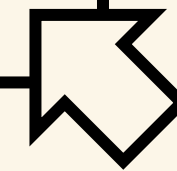
Turns weights into whole
households, each a copy
of a real survey household

38,092

survey households used as the seed

~2.7 million

synthetic households at full scale



118,000

persons in the 1% simulation sample

Placing daily activities

Home

Random OSM building of
40–400 m² inside the
household's ward

Secondary

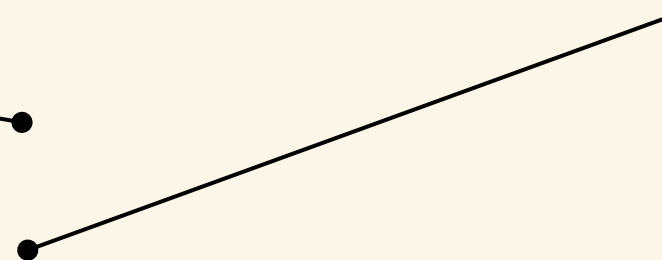
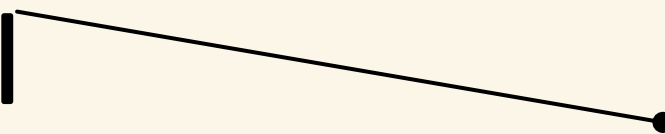
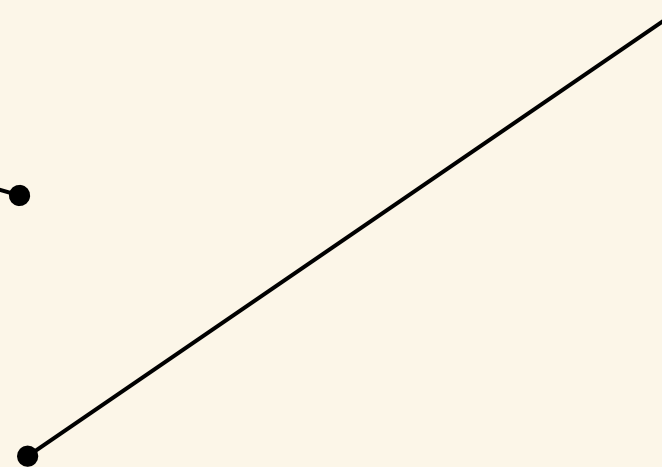
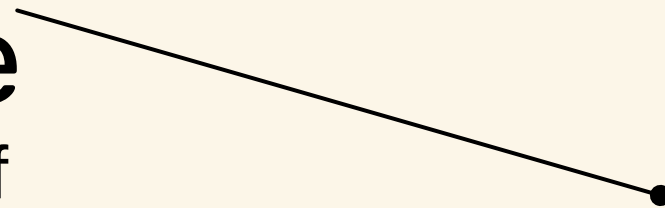
Shopping and leisure
placed with the CARLA
algorithm

Work, school

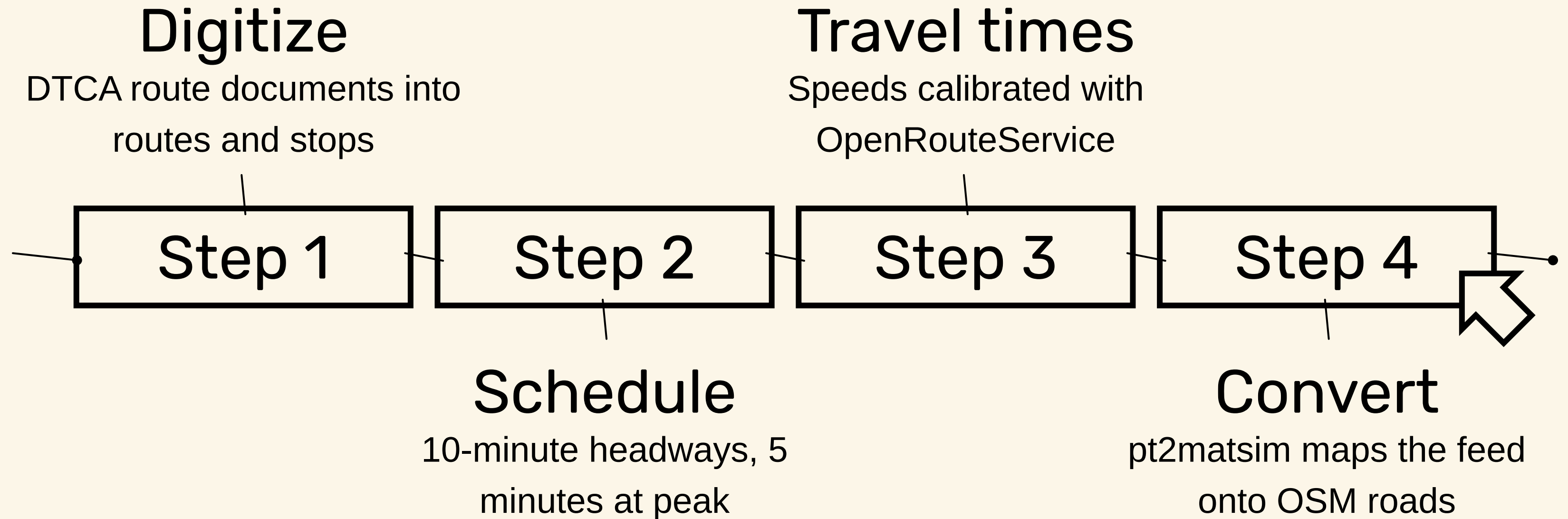
Target distance drawn from
the survey; closest match
found with a k-d tree

~91%

Education locations
assigned after adding OSM
amenity points

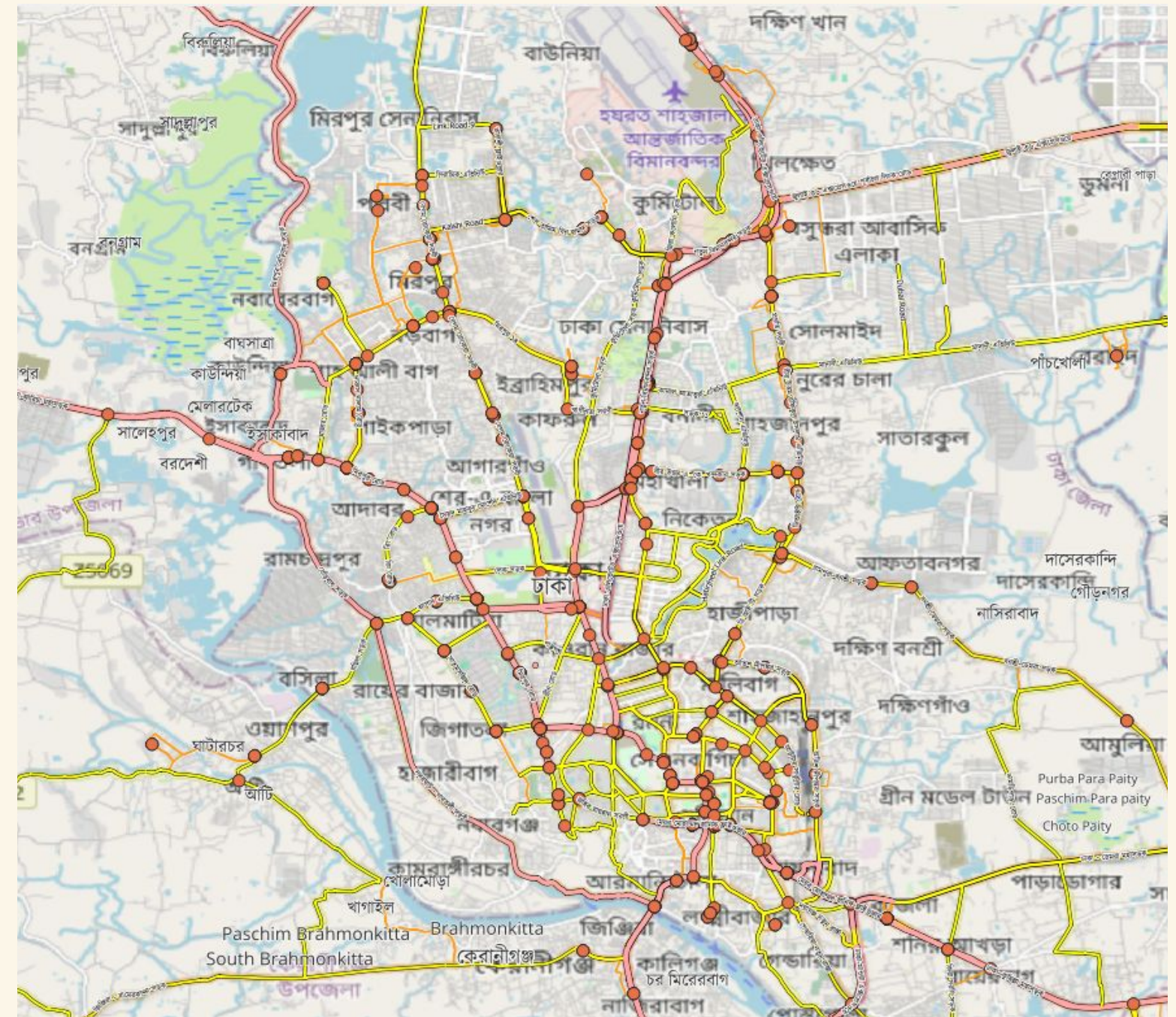


Building the transit feed



The reconstructed bus network

77 routes · 970 stops · 226,602 road links
Basemap © OpenStreetMap contributors



Validation

Age

Within 0.3 pp of census in
every cohort

Households

Within 0.15 pp of census

Chains

Same rank order as the
survey

Distances

Match the survey by
purpose

Employment

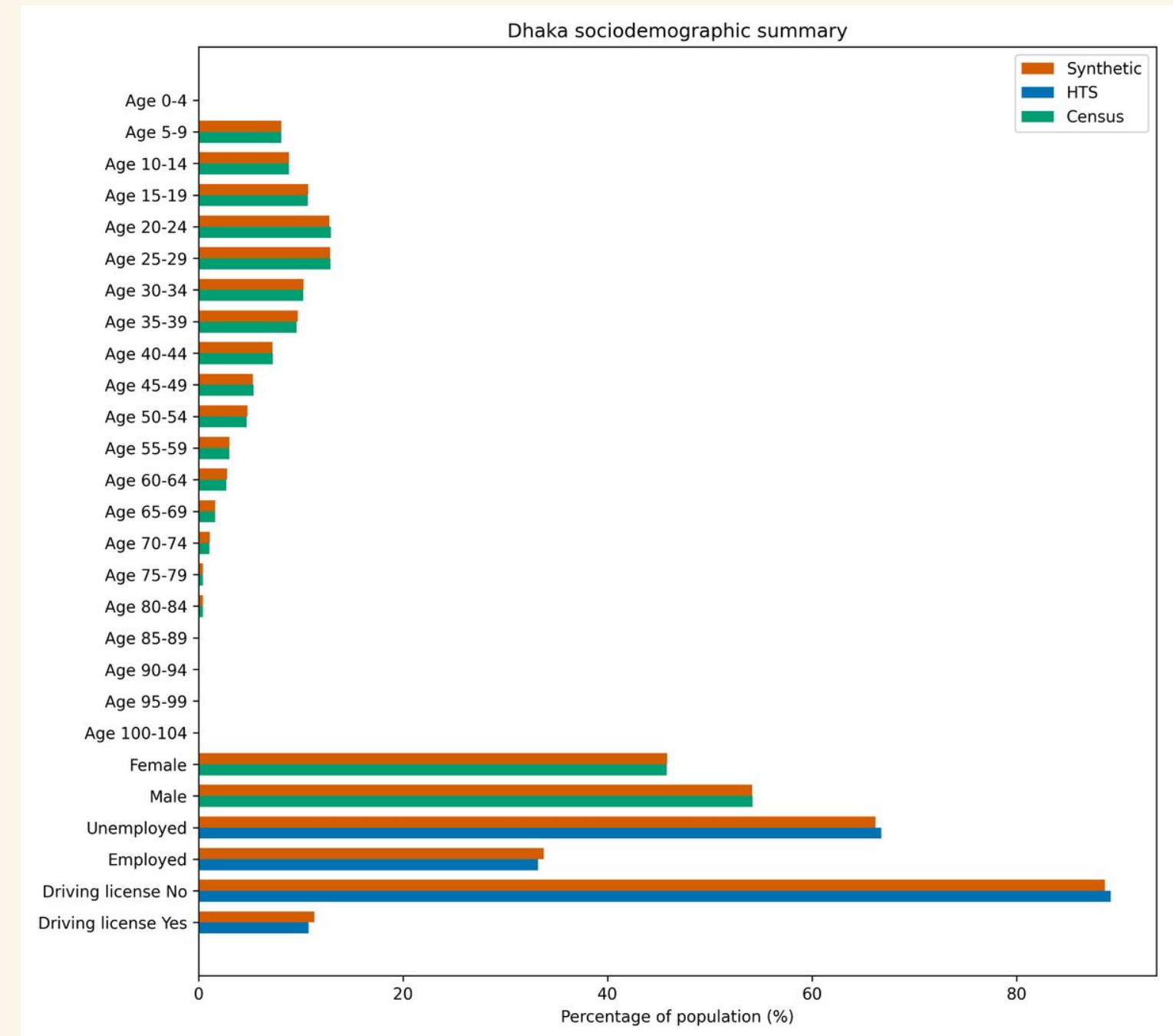
Matches the survey

Licences

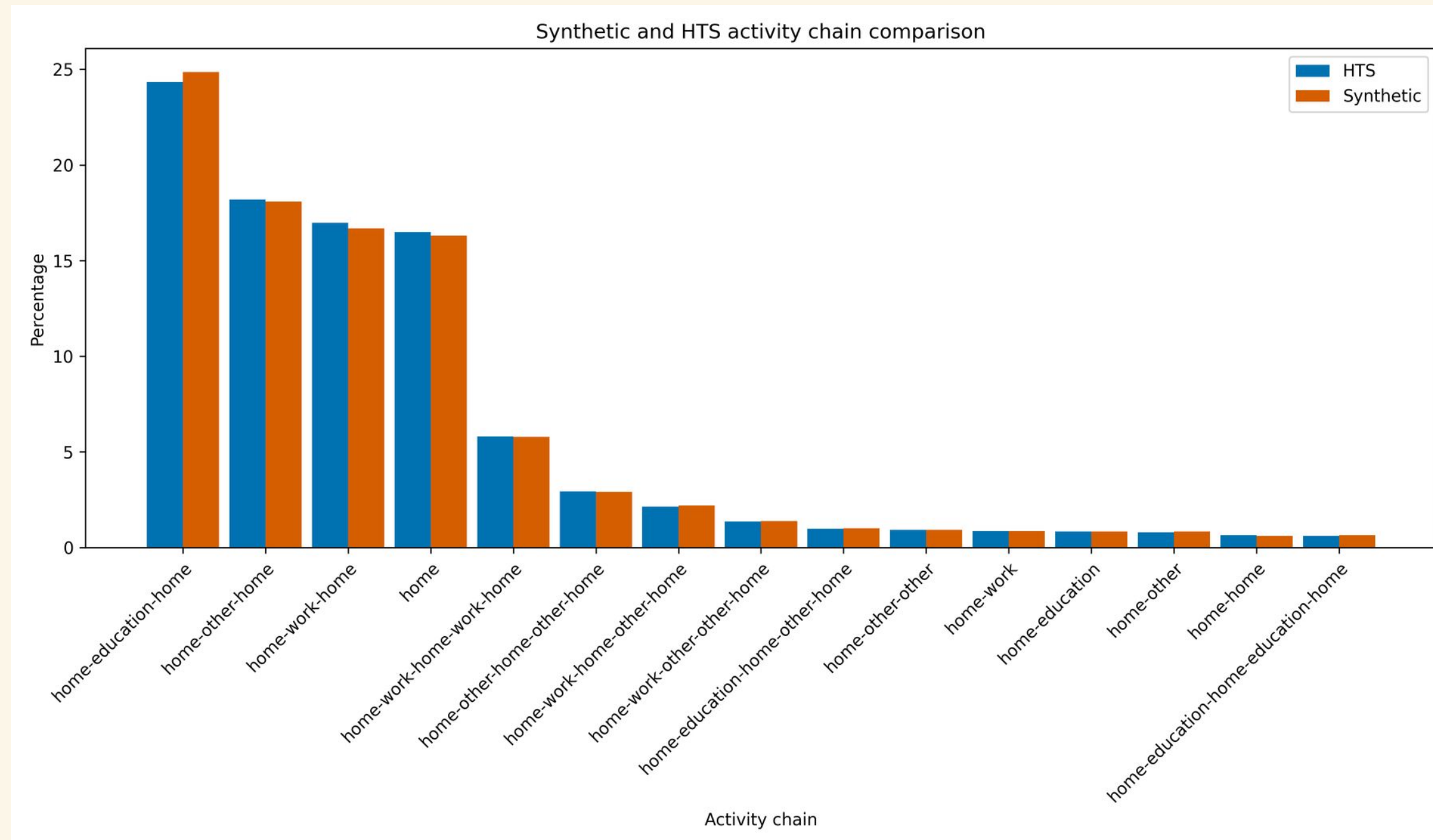
Matches the survey

Population profile matches

Age and sex match the census. Employment
and driving licences match the survey.



Activity chains match

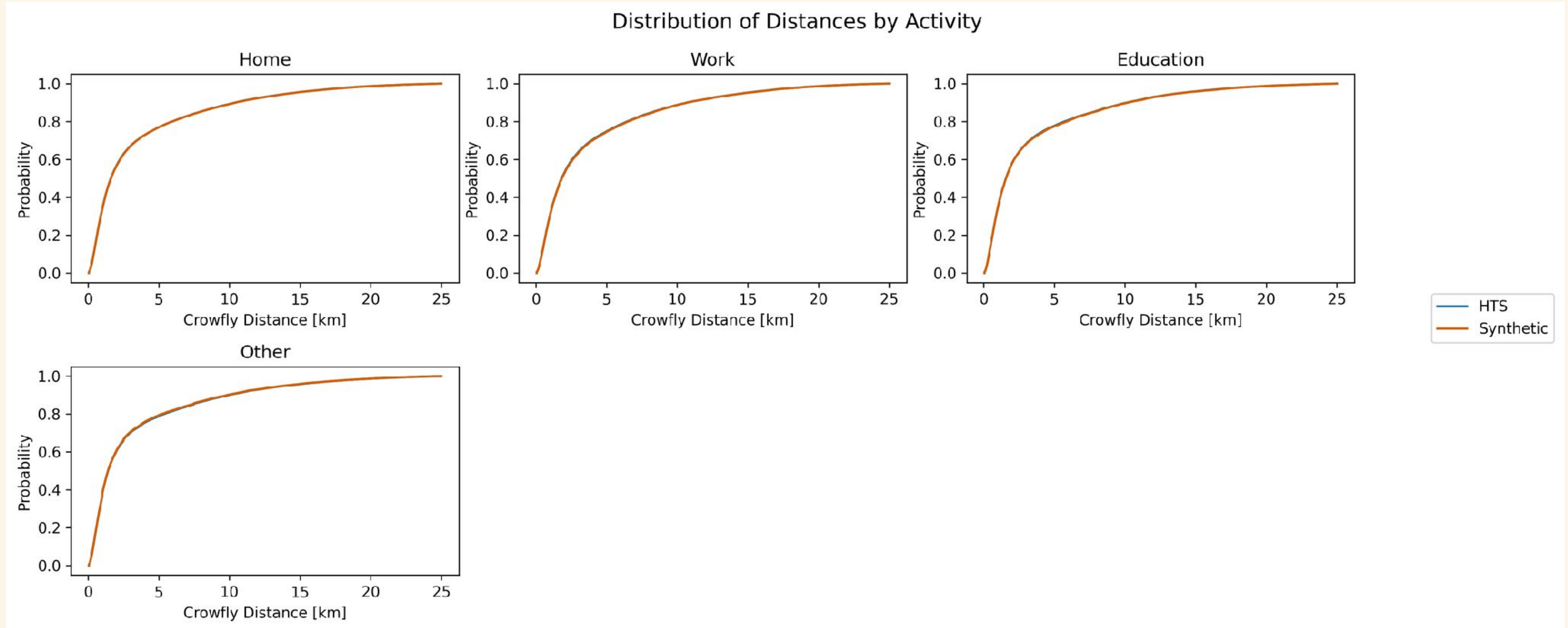


Top 4

chains cover about
three-quarters of people

Each synthetic person
inherits a chain from a
matched survey record

Trip distances match



Limitations

Suburban zones

Savar and Keraniganj rely on survey expansion factors, with no ward-level census

Household diversity

Only 29 single-person and 3,313 five-plus households in the donor pool

Pre-metro baseline

Input data predates Metro Rail MRT Line 6 (opened December 2022)

Next steps

- Add MRT Line 6 to measure modal shift and congestion
- Estimate a discrete mode-choice model
- Apply the pipeline to other Global South megacities

Data scarcity slows agent-based modeling, but need not stop it.

–Key takeaway

Thanks!

Questions?

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